

Sumit Nayek

Project Research Scientist
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Summary

Data professional experienced in statistical analysis and uncovering insights from complex datasets, identifying key patterns, diagnosing root causes for events that drive business decisions. A collaborative problem solver who transforms data into actionable strategies.

Academic Qualifications

Year	Degree	Institute	CGPA
2020 - 2022	M.sc in Statistics	Aliah University, Kolkata	7.57/ 10
2017-2020	B.sc in Statistics	Midnapore College, West bengal	7.74/ 10

Work Experience

- **Project Research Scientist** (National Institute of Technical Teacher's Training and Research) *(Sep 2024 – August 2025)*
 - **Covid-19 patient response recommendation system using large language model (ICMR funded project)**
Mentor: Prof. Chandan Chakraborty, Department of Computer Science and Engineering.
 - * The system takes patient-specific data (e.g., symptoms, medical history, lab results) as input.
 - * A machine learning (ML) model processes this data and generates a predicted classification .
 - * The Large Language Model (LLM) interprets the output of the ML model and formulates a clinically relevant patient-specific recommendation.
 - **COVID 19 Disease Risk Prediction and Diagnostic Classification using Statistical and Machine Learning Techniques (ICMR funded project)**
 - * Designed a machine learning model (e.g., logistic regression, random forest) that processes patient-reported symptoms and other parameters (fever, cough, etc.) to predict infection likelihood.
 - * Validated predictive models with statistical measures (accuracy, precision, recall, AUC-ROC) to ensure reliability.
- **Junior Research Fellow** (National Institute of Technical Teacher's Training and Research) *(Dec 2022 – Aug 2024)*
 - **COVID Disease Data collection, Pre-processing, Statistical Analysis and Visualization.(ICMR funded project)**
Mentor: Prof. Chandan Chakraborty, Department of Computer Science and Engineering.
 - * Processed the raw data from the json files using powerBI and Dax Studio.
 - * The processed data set was analyzed using multiple statistical testing methods to derive meaningful insights.
 - * Evaluated the demographic component trend associated with covid-19 and identification of host site zones.

Technical Skills

- **Analytical:** Exploratory Data Analysis, Complex Data Handling, Data Mining and Visualization, Statistical Methodology, Time Series Analysis, Machine Learning, Neural Network, Natural Language Processing.
- **Programming Languages:** R, Python, C, SQL
- **Software and Libraries:** PowerBI, Tableau, GitHub, SPSS, Microsoft Excel, Tensorflow, Keras ,Sklearn, Seaborn, Matplotlib, Numpy, Pandas.

Personal Projects

- **Cold Mail Generator using LLM** *(Feb 2025 - Jun 2025)*
 - Built an automated cold emailing tool using Llama-3-2B to generate personalized emails from job descriptions, reducing manual effort by 90 percent.
 - Hugging face API keys have been used to access the large language model model.
 - Designed a one-click email generator (Streamlit UI) that combines job descriptions with user profiles to create tailored outreach emails.
- **Psoriasis skin disease classification using transfer learning approach.** *(Jan 2024 - July 2024)*
Mentor: Prof. Chandan Chakraborty,
 - Processed and augmented dermatological datasets (ISIC/HAM10000) using OpenCV and TensorFlow ImageDataGenerator to address limited training samples.
 - Developed a transfer learning-based classifier using MobileNetV2 (pre-trained on ImageNet) to detect psoriasis from skin lesion images.
 - Fine-tuned the model by replacing top layers and optimizing with AdamW optimizer, achieving a F1-score of 0.98 on imbalanced test data.

- **Heart disease risk prediction using Machine learning models**

(Feb 2022 - July 2022)

Mentor: Aliah University(Dr. Biman Chakraborty).

- Data was collected from the open source repository.
- Performed exploratory data analysis (EDA) and feature engineering to identify key risk factors (e.g., age, resting BP) using correlation matrices and visualizations.
- Addressed class imbalance in patient data using SMOTE (Synthetic Minority Oversampling Technique), improving minority class recall by 35 percent compared to random oversampling and reducing model bias toward majority classes.
- Developed a machine learning model (Decision Tree, Bayes Classifier etc.) to predict heart disease risk using clinical parameters (e.g., cholesterol, blood pressure.)

- **Cluster Analysis with Iris Dataset**

(Sep 2021 - Nov 2021)

- K-mean with best choice of k, Gap-stat, lumping and splitting error.
- Hierarchical clustering using complete and single linkage.

Publications

- Chakraborty, C., Achar, U., Nayek, S., Achar, A., Mukherjee, R. (2024). CAD-PsorNet: deep transfer learning for computer-assisted diagnosis of skin psoriasis. Scientific Reports, 14(1), 26557. [Link](#)
- Nayek, S., Chaulia, P. P., Mukherjee, R.,Chakraborty, C. (2024, March). Exploring Deep Transfer Learning for Skin Psoriasis Detection Using Digital Imaging. In International Ethical Hacking Conference (pp. 81-90). Singapore: Springer Nature Singapore.[Link](#)
- Sumit Nayek, Prof. Chandan Chakraborty, Prof. Debi Prasad Mishra and Prof. Rashmi Mukherjee,”Statistical Data Visualization and Machine Learning framework for Diagnostic Prediction of COVID-19 Disease Using Clinical Information”, In 16th International Conference on Computing, Communication and Networking Technologies (ICCCNT 2025). communicated

Certifications

- Oracle Fusion AI Agent Studio Certified Foundation Associate [Link](#)
- Hugging face NLP Course on LLM
- Oracle AI Foundation Certificate on the Oracle Cloud Infrastructure platform [Link](#)
- Advance SQL certification from Hacker Rank.
- Introduction to Data analysis certification from Analytics Vidya.
- Advance Machine Learning certification from Kaggle.
- Certificate on Linear Algebra for Machine Learning from DeepLearning.AI. [Link](#)